CALPHADHEACalc: Governing Equations and Variable Reference

Thermodynamic Formulations, Physical Models, and Implementation Notes

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Abstract — This reference manual documents the thermodynamic, mechanical, kinetic, and transport equations implemented in the CALPHADHEACalc software. For each model, we state the governing relationship, the physical meaning and dimensions of its variables, the corresponding software subroutines, and the supporting literature citations. The formulations cover CALPHAD equilibrium thermodynamics, empirical solid-solution criteria, soft-sphere lattice distortion, yield strength superposition, transport properties across the solidus-liquidus window, high-temperature oxidation and electrochemical corrosion kinetics, and kinematic X-ray diffraction.

1. CALPHAD Solution Thermodynamics and Phase Equilibria

Primary Modules: HEAThermoEngine.m, HEATDBEngine.m

1.1 Molar Gibbs Free Energy of Phase φ

Implementation: Evaluated in `HEAThermoEngine.m` as the thermodynamic objective function minimized during temperature and composition stepping.

$$G_m^{\phi}(T, \mathbf{c}) = {}^{\text{ref}}G^{\phi}(T) + {}^{\text{id}}G^{\phi}(T) + {}^{\text{E}}G^{\phi}(T) + {}^{\text{mag}}G^{\phi}(T) \tag{1}$$

Variables:

- $G_m^{\phi}(T, \mathbf{c})$: Molar Gibbs free energy of phase ϕ at absolute temperature T [$\mathrm{J \cdot mol^{-1}}$].
- T: Absolute temperature [K].
- $\mathbf{c} = (c_1, c_2, \ldots, c_n)$: Mole fraction vector of constituent alloy components [dimensionless, $\sum_{i=1}^{n} c_i = 1$].
- ${}^{\text{ref}}G^{\phi}(T)$: Reference energy of the pure components in the structural state of phase ϕ [$\mathrm{J \cdot mol^{-1}}$].
- ${}^{\text{id}}G^{\phi}(T)$: Ideal mixing contribution [$\mathrm{J \cdot mol^{-1}}$].
- ${}^{\text{E}}G^{\phi}(T)$: Excess Gibbs free energy accounting for atomic pair interactions [$\mathrm{J \cdot mol^{-1}}$].
- ${}^{\text{mag}}G^{\phi}(T)$: Magnetic ordering contribution from the Inden-Hillert-Jarl formulation [$\mathrm{J \cdot mol^{-1}}$].

1.2 Reference Energy from SGTE Unary Lattice Stabilities

Implementation: Evaluated in `HEATDBEngine.m` using polynomial coefficients from the SGTE unary database (Dinsdale, 1991).

$${}^{\text{ref}}G^{\phi}(T) = \sum_{i=1}^{n} c_i \, {}^{\circ}G_i^{\phi}(T) \tag{2}$$

$${}^{\circ}G_i^{\phi}(T) - H_i^{\text{SER}} = a_i + b_i T + c_i T \ln T + d_i T^2 + e_i T^{-1} + f_i T^3 \tag{3}$$

Variables:

- ${}^{\circ}G_i^{\phi}(T)$: Standard molar Gibbs energy of pure element i in phase ϕ [$\mathrm{J \cdot mol^{-1}}$].
- H_i^{SER}: Enthalpy of pure element i in its Standard Element Reference (SER) state at $298.15\ \mathrm{K}$ and $10^5\ \mathrm{Pa}$ [$\mathrm{J \cdot mol^{-1}}$].
- $a_i, b_i, c_i, d_i, e_i, f_i$: Temperature-dependent polynomial coefficients.
- c_i: Mole fraction of element i [$\mathrm{mol/mol}$].

1.3 Ideal Solution Configurational Entropy

Implementation: Evaluated in `HEAThermoEngine.m` for random mixing on the crystal sublattice.

$${}^{\text{id}}G^{\phi}(T) = RT \sum_{i=1}^{n} c_i \ln c_i \tag{4}$$

Variables:

- R: Universal gas constant, $8.314462\,\mathrm{J \cdot mol^{-1} \cdot K^{-1}}$.
- T: Absolute temperature [K].
- c_i: Mole fraction of component i ($0 < c_i \leq 1$).

1.4 Redlich-Kister Multicomponent Excess Gibbs Energy

Implementation: Evaluated in `HEAThermoEngine.m` using binary and ternary Redlich-Kister interaction coefficients with Muggianu geometric extrapolation.

$$^{\mathrm{E}}G^{\phi}(T) = \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} c_i c_j \sum_{k=0}^{p} L_{ij}^{\phi,(k)}(T)\,(c_i - c_j)^k, \quad L_{ij}^{\phi,(k)}(T) = a_k + b_k T \tag{5}$$

Variables:

- $L_{ij}^{\phi,(k)}(T)$: k-th order binary interaction parameter between elements i and j in phase ϕ [$\mathrm{J\cdot mol^{-1}}$].
- a_k: Enthalpic interaction coefficient [$\mathrm{J\cdot mol^{-1}}$].
- b_k: Entropic interaction coefficient [$\mathrm{J\cdot mol^{-1}\cdot K^{-1}}$].
- p: Expansion order ($p = 0$ for regular solutions; $p \geq 1$ for sub-regular solutions).

1.5 Chemical Potential Balance and Multiphase Equilibrium

Implementation: Solved in `HEAThermoEngine.m` using a constrained Newton-Raphson scheme to match chemical potentials under mass conservation.

$$\mu_i^{\alpha}(T, \mathbf{x}^{\alpha}) = \mu_i^{\beta}(T, \mathbf{x}^{\beta}) = \cdots = \mu_i^{\nu}(T, \mathbf{x}^{\nu}) \quad (\forall i = 1, \ldots, n) \tag{6}$$

$$\sum_{\phi=1}^{\Phi} f_{\phi} x_i^{\phi} = c_i^0, \quad \sum_{\phi=1}^{\Phi} f_{\phi} = 1 \tag{7}$$

Variables:

- μ_i^{ϕ}: Chemical potential of component i in phase ϕ [$\mathrm{J\cdot mol^{-1}}$], where $\mu_i^{\phi} = (\partial G^{\phi}/\partial n_i)_{T,P,n_{j\neq i}}$.
- f_{ϕ}: Molar phase fraction of phase ϕ ($0 \leq f_{\phi} \leq 1$).
- $\mathbf{x}^{\phi} = (x_1^{\phi}, \ldots, x_n^{\phi})$: Composition of phase ϕ [$\mathrm{mol/mol}$].
- c_i^0: Overall alloy mole fraction of component i.
- $\alpha, \beta, \ldots, \nu$: Equilibrium phases (e.g. FCC_A1, BCC_A2, B2, Liquid, Sigma).

2. Empirical Solid-Solution Criteria and Solidification

Primary Module: HEAThermoEngine.m

2.1 Miedema Enthalpy of Mixing and Configurational Entropy

Implementation: Evaluated in `HEAThermoEngine.m` using Miedema regular-solution interaction enthalpies.

$$\Delta H_{\mathrm{mix}} = \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} 4\,\Omega_{ij}\,c_i\,c_j, \quad \Delta S_{\mathrm{mix}} = -R\sum_{i=1}^{n} c_i \ln c_i \tag{8}$$

Variables:

- ΔH_{mix}: Molar enthalpy of mixing [$\mathrm{kJ\cdot mol^{-1}}$].
- Ω_{ij}: Equiatomic binary mixing enthalpy between elements i and j [$\mathrm{kJ\cdot mol^{-1}}$].
- ΔS_{mix}: Configurational entropy of mixing [$\mathrm{J\cdot mol^{-1}\cdot K^{-1}}$].
- c_i, c_j: Atomic fractions of components i and j.

2.2 Atomic Size Mismatch (δ) and Electronegativity Difference (Δχ)

Implementation: Evaluated in `HEAThermoEngine.m` to assess Hume-Rothery topological strain criteria ($\delta \leq 6.6\%$) and chemical electronegativity differences.

$$\delta = \sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{r_i}{\bar{r}}\right)^2} \times 100\%, \quad \bar{r} = \sum_{i=1}^{n} c_i r_i \tag{9}$$

$$\Delta\chi = \sqrt{\sum_{i=1}^{n} c_i (\chi_i - \bar{\chi})^2}, \quad \bar{\chi} = \sum_{i=1}^{n} c_i \chi_i \tag{10}$$

Variables:

- δ: Atomic size difference parameter [$\%$].
- r_i: Goldschmidt atomic radius of element i [$\mathrm{\AA}$].
- $\bar{r}$: Average atomic radius of the alloy system [$\mathrm{\AA}$].
- χ_i: Pauling electronegativity of element i.
- $\Delta\chi$: Root-mean-square electronegativity mismatch.

2.3 Yang-Zhang Thermodynamic Parameter (Ω)

Implementation: Evaluated in `HEAThermoEngine.m` using the Yang-Zhang criterion ($\Omega \geq 1.1$ and $\delta \leq 6.6\%$).

$$\Omega = \frac{T_m \, \Delta S_{\text{mix}}}{|\Delta H_{\text{mix}}|}, \quad T_m = \sum_{i=1}^{n} c_i T_{m,i} \tag{11}$$

Variables:

- Ω: Thermodynamic stability parameter [dimensionless].
- T_m: Rule-of-mixtures melting temperature [K].
- $T_{m,i}$: Melting point of pure component i [K].

2.4 Guo Valence Electron Concentration (VEC) Criteria

Implementation: Evaluated in `HEAThermoEngine.m` following the phase boundary criteria of Guo et al.: $\mathrm{VEC} \geq 8.0$ favors FCC; $\mathrm{VEC} < 6.87$ favors BCC; $6.87 \leq \mathrm{VEC} < 8.0$ indicates mixed FCC+BCC.

$$\mathrm{VEC} = \sum_{i=1}^{n} c_i \, \mathrm{VEC}_i \tag{12}$$

Variables:

- VEC: Valence electron concentration of the alloy [electrons/atom].
- VEC_i: Valence electron count of element i.

2.5 Scheil-Gulliver Solidification and Homogenization Kinetics

Implementation: Evaluated in `HEAThermoEngine.m` to estimate solute segregation under rapid cooling and calculate homogenization heat treatment conditions.

$$T_{\text{ScheilSolidus}} = T_{\text{Solidus}} - 0.45 \cdot (T_{\text{Liquidus}} - T_{\text{Solidus}}) \cdot (1 + 0.05 \cdot \delta) \tag{13}$$

$$I_{\text{seg}} = 0.25 + 0.06 \cdot \delta, \quad T_{\text{homog}} = \min(T_{\text{Solidus}} - 50\ \mathrm{K},\, 0.88 \cdot T_{\text{Solidus}}), \quad \tau_{\text{homog}} = 10.0 \cdot \left(\frac{I_{\text{seg}}}{0.35}\right)^2 \ \text{hours} \tag{14}$$

Variables:

- $T_{\text{ScheilSolidus}}$: Non-equilibrium solidus temperature [K].
- $T_{\text{Solidus}}, T_{\text{Liquidus}}$: Equilibrium solidus and liquidus temperatures [K].
- I_{seg}: Segregation severity index.
- T_{homog}: Annealing homogenization temperature [K].
- τ_{homog}: Isothermal soaking time [hours].

3. Lattice Distortion and Dislocation Mechanics

Primary Module: HEALatticeDistortionEngine.m

3.1 Effective Relaxed Atomic Radius

Implementation: Evaluated in `HEALatticeDistortionEngine.m` following the elastic soft-sphere model of He and Yang (2018).

$$r_i^* = r_i \left(1 - \beta \frac{K_i - \bar{K}}{\bar{K}}\right), \quad \beta = \frac{1+\nu}{3(1-\nu)}, \quad \bar{K} = \sum_{i=1}^{n} c_i K_i \tag{15}$$

Variables:

- r_i^*: Effective relaxed atomic radius of component i in the solid solution [$\mathrm{\AA}$].
- r_i: Unperturbed metallic radius [$\mathrm{\AA}$].
- $K_i, \bar{K}$: Bulk modulus of component i and composition-weighted average bulk modulus [GPa].
- ν: Average Poisson's ratio ($\approx 0.30 - 0.33$).
- β: Elastic stress accommodation factor.

3.2 Root-Mean-Square Lattice Strain

Implementation: Evaluated in `HEALatticeDistortionEngine.m` to quantify static atomic displacements.

$$\varepsilon_{\text{RMS}} = \sqrt{\sum_{i=1}^{n} c_i \left(\frac{r_i^* - \bar{r}^*}{\bar{r}^*}\right)^2}, \quad \bar{r}^* = \sum_{i=1}^{n} c_i r_i^* \tag{16}$$

Variables:

- ε_{RMS}: Root-mean-square static displacement strain [dimensionless].
- $\bar{r}^*$: Mean effective atomic radius [$\mathrm{\AA}$].

3.3 Theoretical Shear Strain and Collective Solute Pinning

Implementation: Evaluated in `HEALatticeDistortionEngine.m` to compute theoretical shear resistance and the Labusch solute pinning increment.

$$\bar{\gamma}_{\text{th}} = \gamma_0 \left(1 - \alpha_{\text{dist}} \varepsilon_{\text{RMS}}\right) \tag{17}$$

$$\tau_c = \tau_{\text{Peierls}} + \Delta\tau_{\text{Labusch}}, \quad \Delta\tau_{\text{Labusch}} = A_L \cdot G \cdot \varepsilon_{\text{RMS}}^{4/3} \cdot c_i^{2/3} \tag{18}$$

Variables:

- $\bar{\gamma}_{\text{th}}$: Theoretical plastic shear strain limit.
- γ_0: Ideal shear strain of unperturbed lattice (≈ 0.11).
- α_{dist}: Lattice distortion coupling coefficient (≈ 2.8).
- τ_c: Critical resolved shear stress [MPa].
- τ_{Peierls}: Intrinsic lattice friction stress [MPa].
- $\Delta\tau_{\text{Labusch}}$: Solid-solution strengthening increment [MPa].
- G: Shear modulus [GPa].
- A_L: Dimensionless scaling factor.

3.4 Dislocation Activation Volume

Implementation: Evaluated in `HEALatticeDistortionEngine.m` for thermally activated dislocation glide.

$$V^* = b \cdot d_w \cdot l_s = -\frac{\partial \Delta G^*}{\partial \tau} \tag{19}$$

Variables:

- V^*: Dislocation activation volume [m^3 or in units of b^3].
- b: Magnitude of Burgers vector [nm] ($\approx 0.255\ \text{nm}$ for FCC, $\approx 0.286\ \text{nm}$ for BCC).
- d_w: Obstacle barrier width [nm].
- l_s: Average spacing between pinning obstacles [nm].

4. Mechanical Strengthening and Constitutive Relations

Primary Module: HEAMechanicalEngine.m

4.1 Yield Strength Superposition

Implementation: Evaluated in `HEAMechanicalEngine.m` to calculate 0.2% offset engineering tensile yield strength.

$$\sigma_y = \sigma_0 + \Delta\sigma_{\text{ss}} + \Delta\sigma_{\text{HP}} + \Delta\sigma_{\text{ppt}} \tag{20}$$

Variables:

- σ_y: Engineering yield strength [MPa].
- σ_0: Lattice friction stress [MPa] ($\sim 45\ \text{MPa}$ for FCC; $\sim 650 - 950\ \text{MPa}$ for BCC).
- $\Delta\sigma_{\text{ss}}$: Solid-solution strengthening contribution [MPa].
- $\Delta\sigma_{\text{HP}}$: Hall-Petch grain boundary strengthening [MPa].
- $\Delta\sigma_{\text{ppt}}$: Precipitation strengthening contribution [MPa].

4.2 Solid Solution Strengthening: Varvenne-Curtin (FCC) and Maresca-Curtin (BCC)

Implementation: Evaluated in `HEAMechanicalEngine.m` based on the random alloy strengthening theories of Varvenne et al. (2016) and Maresca and Curtin (2020).

$$\Delta\sigma_{\text{ss}}^{\text{FCC}} = 0.0405 \cdot M \cdot \mu \left(\frac{1+\nu}{1-\nu}\right)^{4/3} \left[\sum_{i=1}^{n} c_i \left(\frac{\Delta V_i}{V}\right)^2\right]^{2/3} \tag{21}$$

$$\Delta\sigma_{\text{ss}}^{\text{BCC}} = M \cdot \left[\tau_{\text{kink}} + \alpha_k \cdot \mu \cdot \Delta\varepsilon_{\text{size}}^{4/3}\right] \tag{22}$$

Variables:

- M: Taylor orientation factor ($M = 3.06$ for FCC; $M = 2.73$ for BCC).
- μ: Shear modulus [GPa].
- ν: Poisson's ratio.
- $\Delta V_i / V$: Normalized misfit volume of solute element i.
- τ_{kink}: Stress threshold for screw dislocation kink-pair nucleation [MPa].

4.3 Hall-Petch Hardening and High-Temperature Softening

Implementation: Evaluated in `HEAMechanicalEngine.m` for grain-size hardening and thermal softening up to the melting point.

$$\Delta\sigma_{\mathrm{HP}} = \frac{k_{\mathrm{HP}}}{\sqrt{d}}, \quad \sigma_y(T) = \sigma_y(298\ \mathrm{K}) \left[1 - \left(\frac{T - 298}{T_m - 298}\right)^m\right] \tag{23}$$

Variables:

- k_{HP}: Hall-Petch coefficient [$\mathrm{MPa} \cdot \mu\mathrm{m}^{1/2}$] ($494\ \mathrm{MPa} \cdot \mu\mathrm{m}^{1/2}$ for CoCrFeMnNi).
- d: Mean grain diameter [$\mu\mathrm{m}$] ($30\ \mu\mathrm{m}$ for standard reference state).
- T_m: Solidus temperature [K].
- m: Thermal softening exponent ($0.65 - 0.85$).

4.4 Hardness, True Flow Stress, and Wear Loss

Implementation: Evaluated in `HEAMechanicalEngine.m` for Vickers microhardness, Hollomon work-hardening, Archard wear loss, and cladding dilution.

$$\mathrm{HV} \approx 0.3\,\sigma_y + 60, \quad \sigma_{\mathrm{true}} = K\,\varepsilon_{\mathrm{true}}^n, \quad W_{\mathrm{wear}} = K_{\mathrm{Archard}} \frac{F_N\, s}{H}, \quad \eta = \frac{A_{\mathrm{sub}}}{A_{\mathrm{clad}} + A_{\mathrm{sub}}} \times 100\% \tag{24}$$

Variables:

- HV: Vickers microhardness [$\mathrm{kgf} \cdot \mathrm{mm}^{-2}$].
- K: Hollomon strength coefficient [MPa].
- n: Strain-hardening exponent.
- W_{wear}: Volumetric wear loss [mm^3].
- K_{Archard}: Dimensionless Archard wear coefficient.
- F_N: Normal load [N].
- s: Sliding distance [m].
- η: Substrate dilution fraction [$\%$].

5. Thermophysical Transport Properties

Primary Module: HEATransportEngine.m

5.1 Density, Specific Heat, and Thermal Conductivity

Implementation: Evaluated in `HEATransportEngine.m` for temperature-dependent density, Kopp-Neumann heat capacity, and thermal conductivity.

$$\rho(T) = \frac{\rho(298\ \mathrm{K})}{1 + 3\,\alpha_L (T - 298)}, \quad C_p(T) = \sum_{i=1}^{n} c_i\, C_{p,i}(T), \quad k(T) = \frac{L_0\, T}{\rho_{\mathrm{el}}(T)} + \frac{A_{\mathrm{ph}}}{T} \tag{25}$$

Variables:

- $\rho(T)$: Density at temperature T [$\mathrm{kg} \cdot \mathrm{m}^{-3}$].
- α_L: Linear coefficient of thermal expansion [K^{-1}].
- $C_p(T)$: Molar heat capacity at constant pressure [$\mathrm{J} \cdot \mathrm{mol}^{-1} \cdot \mathrm{K}^{-1}$].
- $k(T)$: Total thermal conductivity [$\mathrm{W} \cdot \mathrm{m}^{-1} \cdot \mathrm{K}^{-1}$].
- L_0: Sommerfeld Lorenz number, $2.443 \times 10^{-8}\ \mathrm{W} \cdot \Omega \cdot \mathrm{K}^{-2}$.
- $\rho_{\mathrm{el}}(T)$: Electrical resistivity [$\Omega \cdot \mathrm{m}$].

5.2 Semi-Solid Viscosity and Laser Absorptivity

Implementation: Evaluated in `HEATransportEngine.m` using the Krieger-Dougherty semi-solid rheology relation and Hagen-Rubens optical absorptivity.

$$\eta(f_s) = \eta_{\mathrm{liq}} \left(1 - \frac{f_s}{f_{s,\mathrm{crit}}}\right)^{-2.5\, f_{s,\mathrm{crit}}}, \quad A_{\mathrm{laser}}(\lambda, T) \approx \sqrt{\frac{8\pi\varepsilon_0 c_0\, \rho_{\mathrm{el}}(T)}{\lambda}} \tag{26}$$

Variables:

- $\eta(f_s)$: Dynamic apparent viscosity in the mushy zone [$\mathrm{Pa} \cdot \mathrm{s}$].
- η_{liq}: Molten liquid viscosity ($\approx 0.003 - 0.006\ \mathrm{Pa} \cdot \mathrm{s}$).
- $f_{s,\mathrm{crit}}$: Critical solid fraction at rheological jamming ($0.50 - 0.60$).
- A_{laser}: Optical surface absorptivity [dimensionless].
- λ: Laser wavelength [m] ($1064\ \mathrm{nm}$ for Nd:YAG).

6. Environmental Kinetics: Oxidation and Corrosion

Primary Modules: HEAOxidationEngine.m, HEACorrosionEngine.m

6.1 Wagner Parabolic Oxidation and Pilling-Bedworth Ratio

Implementation: Evaluated in `HEAOxidationEngine.m` for isothermal oxidation mass gain curves and oxide scale compactness.

$$\left(\frac{\Delta W}{A}\right)^2 = k_p(T)\,t, \quad k_p(T) = k_0 \exp\left(-\frac{Q_{\text{ox}}}{RT}\right), \quad \text{PBR} = \frac{M_{\text{ox}}\,\rho_{\text{met}}}{n_{\text{met}}\,M_{\text{met}}\,\rho_{\text{ox}}} \tag{27}$$

Variables:

- $\Delta W/A$: Mass gain per unit area [$\mathbf{mg \cdot cm^{-2}}$].
- $k_p(T)$: Parabolic rate constant [$\mathbf{mg^2 \cdot cm^{-4} \cdot h^{-1}}$].
- t: Exposure duration [**hours**].
- Q_{ox}: Oxidation activation energy [$\mathbf{kJ \cdot mol^{-1}}$].
- **PBR**: Pilling-Bedworth ratio ($\mathbf{1.0 \leq PBR \leq 2.0}$ indicates protective scale).

6.2 Electrochemical Corrosion Rate (ASTM G102) and PREN

Implementation: Evaluated in `HEACorrosionEngine.m` according to ASTM G102-89 and Butler-Volmer polarization kinetics.

$$\text{CR} = K_{\text{corr}}\,\frac{i_{\text{corr}}\,\text{EW}}{\rho}, \quad \text{EW} = \left[\sum_{i=1}^{n} \frac{w_i\,n_i}{M_i}\right]^{-1}, \quad \text{PREN}_{\text{HEA}} = \text{wt\%Cr} + 3.3(\text{wt\%Mo} + 0.5\,\text{wt\%W}) + 16\,\text{wt\%N} \tag{28}$$

$$i_{\text{net}} = i_{\text{corr}}\left[\exp\left(\frac{2.303(E - E_{\text{corr}})}{\beta_a}\right) - \exp\left(-\frac{2.303(E - E_{\text{corr}})}{\beta_c}\right)\right] \tag{29}$$

Variables:

- **CR**: Annual corrosion penetration rate [$\mathbf{mm \cdot year^{-1}}$].
- K_{corr}: Conversion factor, $\mathbf{0.00327\,mm \cdot g \cdot \mu A^{-1} \cdot cm^{-1} \cdot year^{-1}}$.
- i_{corr}: Corrosion current density [$\mathbf{\mu A \cdot cm^{-2}}$].
- **EW**: Alloy equivalent weight [$\mathbf{g \cdot equivalent^{-1}}$].
- w_i: Mass fraction of element i.
- n_i: Primary valence state in dissolution (e.g. Fe: 2, Cr: 3, Ni: 2, Mo: 4).
- PREN_{HEA}: Pitting resistance equivalent number.
- β_a, β_c: Anodic and cathodic Tafel slopes [$\mathbf{V \cdot decade^{-1}}$].

7. Kinematic Powder X-Ray Diffractometry

Primary Module: HEAXRDEngine.m

7.1 Bragg Diffraction and Structure Factor

Implementation: Evaluated in `HEAXRDEngine.m` to synthesize kinematic X-ray diffraction patterns for cubic and hexagonal phases.

$$\lambda = 2\,d_{hkl}\,\sin\theta_{hkl}, \quad d_{hkl} = \frac{a_0}{\sqrt{h^2 + k^2 + l^2}} \tag{30}$$

$$I_{hkl} \propto |F_{hkl}|^2 \cdot p_{hkl} \cdot \left(\frac{1 + \cos^2(2\theta)}{\sin^2\theta\cos\theta}\right) \cdot \exp(-2B_{\text{DW}}\sin^2\theta/\lambda^2) \tag{31}$$

Variables:

- λ: X-ray wavelength [$\mathbf{\AA}$] ($\mathbf{1.54056\,\AA}$ for **Cu-K**α_1).
- θ_{hkl}: Bragg angle [**degrees**].
- d_{hkl}: Interplanar spacing of (hkl) planes [$\mathbf{\AA}$].
- a_0: Lattice parameter from Vegard's law with soft-sphere relaxation [$\mathbf{\AA}$].
- F_{hkl}: Structure factor for reflection (hkl).
- p_{hkl}: Multiplicity factor of the reflection plane family.
- B_{DW}: Debye-Waller parameter [$\mathbf{\AA^2}$].

7.2 Peak Broadening: Scherrer and Williamson-Hall

Implementation: Evaluated in `HEAXRDEngine.m` to simulate line broadening from finite crystallite size and lattice microstrain.

$$\text{FWHM}_{hkl} = \frac{K_{\text{Scherrer}}\,\lambda}{D_{\text{crystallite}}\cos\theta_{hkl}} + 4\,\varepsilon_{\text{micro}}\tan\theta_{hkl} \tag{32}$$

Variables:

- $\mathbf{FWHM}_{hkl}$: Peak full width at half maximum [$\mathbf{rad}$].
- $K_{\mathbf{Scherrer}}$: Scherrer shape factor ($\approx \mathbf{0.94}$).
- $D_{\mathbf{crystallite}}$: Coherent domain size [$\mathbf{nm}$].
- $\varepsilon_{\mathbf{micro}}$: Root-mean-square microstrain.

8. Application Ranking and COMSOL Export

Primary Modules: HEAApplicationDatabase.m, CALPHADHEACalc.m

8.1 Multi-Criteria Suitability Scoring

Implementation: Evaluated in `HEAApplicationDatabase.m` to rank candidate alloys across 8 engineering application sectors.

$$S_j = \left[\sum_{k=1}^{m} w_k \cdot \phi_k(P_k)\right] \times \prod_{l=1}^{p} \Theta_l, \quad \phi_k(P_k) = \frac{1}{1 + \exp\left(-\kappa_k \cdot \frac{P_k - P_{k,\text{target}}}{\Delta P_k}\right)} \tag{33}$$

Variables:

- S_j: Overall suitability score for sector j ($\mathbf{0} \leq S_j \leq \mathbf{1.0}$).
- w_k: Weighting factor for property k ($\sum w_k = \mathbf{1.0}$).
- $\phi_k(P_k)$: Sigmoidal response function for property $P_k \in [0, 1]$.
- $\Theta_l \in \{0, 1\}$: Hard constraint flag (e.g. elongation $< 5\%$ disqualifies cryogenic applications).

8.2 Continuous Piecewise Property Smoothing for COMSOL

Implementation: Evaluated in `CALPHADHEACalc.m` to export continuous material property functions for finite element simulations.

$$P_{\text{smooth}}(T) = P_{\text{solid}}(T) + [P_{\text{liquid}}(T) - P_{\text{solid}}(T)] \cdot \text{flc2hs}\left(T - T_{\text{Solidus}},\, \Delta T_{\text{smooth}}\right) \tag{34}$$

Variables:

- $P_{\text{smooth}}(T)$: Continuous property function across solid, mushy, and liquid regimes.
- $\text{flc2hs}(x, \Delta x)$: COMSOL smoothed Heaviside function with continuous second derivative.
- ΔT_{smooth}: Smoothing transition temperature width [$\mathbf{K}$].

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